

AWS IoTCore Certification Test Setup

This document captures the steps to setup the AWS IoTCore certification tests with an Avnet RASynBoard.

Load custom DA16600 images to the RASynBoard

If you need to flash a new DA16600 image to the RASynBoard's DA16600, follow the [instructions here](#).

- Note: The documentation link above also details where to connect a USB to TTL debug cable to the RASynBoard to view DA16600 debug output

Prepare your RASynBoard with the AWS test RA6M4 application

There have been changes implemented in the RASynBoard RA6M4 application to pass the tests that have not been released yet. For example I added AT commands to set the MQTT version to 3.1.1 and some other minor things required to pass the tests.

Link to files: [Link](#)

When you follow the [Link](#) you'll find . . .

- Prebuilt RA6M4 Image: **aws_certification.srec**
- MicroSD Card files to load onto the RASynBoard MicroSD Card: **RASynMicroSdCardFiles.zip**
- RA6M4 source code that generated the *.srec file: **RASynBoardSourceCode.zip**

To load the image and microSD card files follow the [instructions here](#)

- **Please use the files downloaded above, not a published release. Use the image from the link above**

To build the source code and debug the RASynBoard application follow the [instructions here](#).

- All the AT commands and state machine that implements the AWS IoTCore connection can be found in **/src/iotc_thread_entry.c** file.

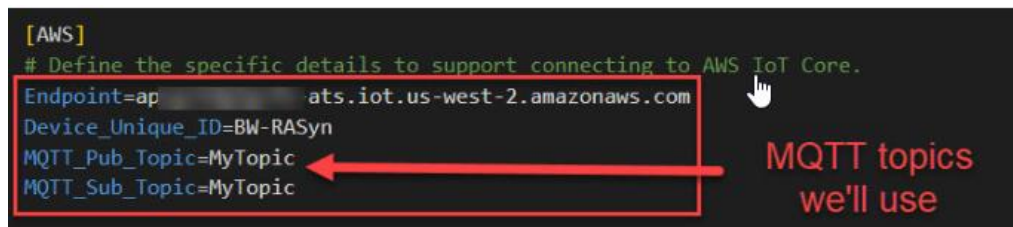
Configure the WiFi on the DA16600

Instructions for configuring the WiFi network can be [found here](#)

Create an AWS account and create a device to test with

Please follow along with the documentation [here](#).

- Make sure you can connect the device to AWS IoT Core before moving on, this will verify that the certificates and RASynBoard configuration are correct
 - When the application connects to AWS IoT Core it will automatically send multiple telemetry messages. If your publish and subscribe topics are the same, then AWS IoT Core will receive the telemetry messages and then send them back to the device.
 - All these messages will be show in debug if you connected a USB to TTL debug cable
- Make a note of the device name that you created, for example **"BW-RASyn"**
- Make a note of the **MQTT_Pub_Topic** and **MQTT_Sub_Topics** you set in the RASynBoard config.ini file. For example . . .
- We'll use these details when we setup the certification test



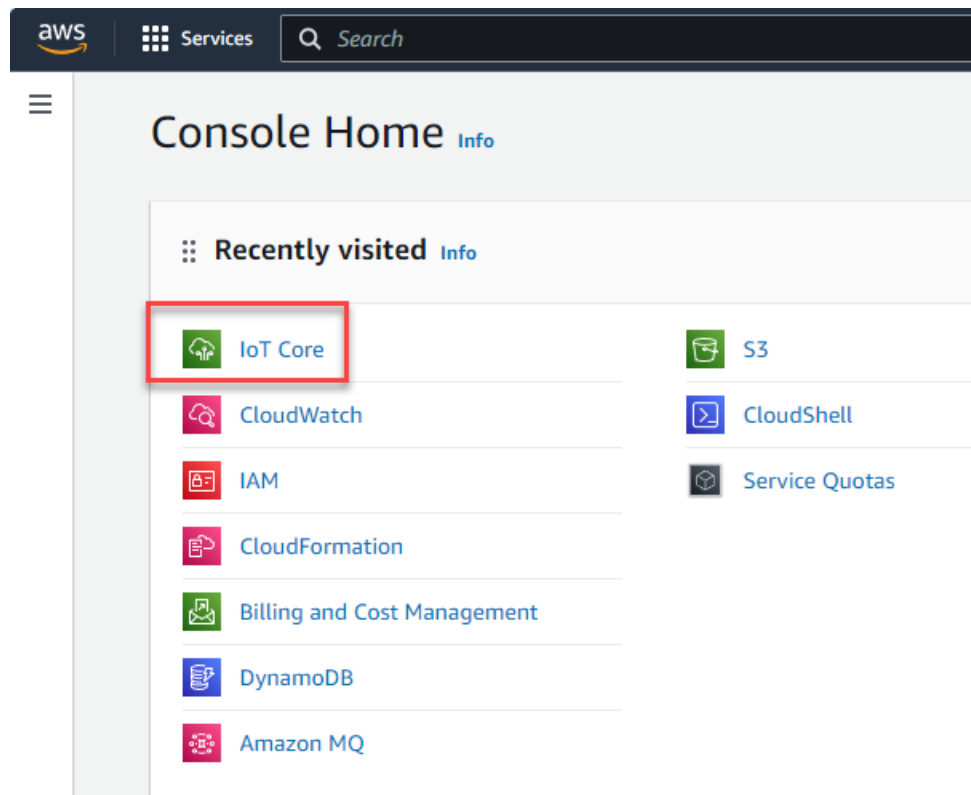
```
[AWS]
# Define the specific details to support connecting to AWS IoT Core.
Endpoint=ap[REDACTED]ats.iot.us-west-2.amazonaws.com
Device_Unique_ID=BW-RASyn
MQTT_Pub_Topic=MyTopic
MQTT_Sub_Topic=MyTopic
```

MQTT topics we'll use

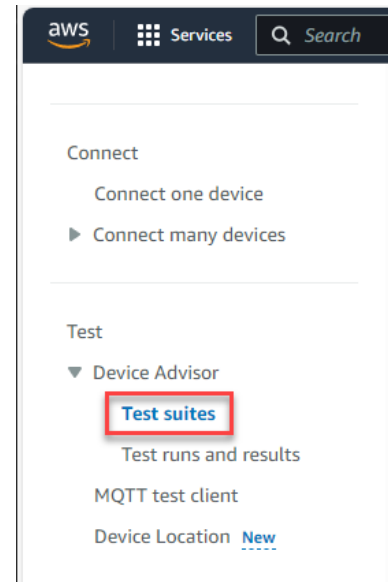
Create a Device Advisor Test Suite

AWS provides a test suite that devices must pass to be included in the [AWS Certified Device Catalog](#). By listing devices in this catalog, we get free exposure to potential customers looking for a platform for new IoT Devices. To create a new test suite . . .

1. Log into the [AWS console](#) using the account credentials setup above
2. Select **"IoT Core"**



3. From the menu at the left select **Test**→**Device Advisor**→**Test Suites**



4. Select "Create qualification test suite"

Test

▼ Device Advisor

Test suites

Test runs and results

MQTT test client

Device Location [New](#)

Manage



AWS IoT Core qualification test suite

Qualify your device for inclusion in the AWS Partner Device Catalog.

Create qualification test suite

5. The Avnet RASynBoard OOB application is not having any issues passing the MQTT test, so let's just create a custom test suite to validate that the DA16600 TLS implementation passes the tests.
6. Select "Custom test suite" and "MQTT 3.1.1", then **"Next"**

Choose test suite type [Info](#)

☐ AWS IoT Core qualification test suite
Qualify a device for the AWS Device Partner Catalog.

☐ Long duration test suite
Monitor your device behavior when tested for a long duration with multiple test scenarios.

☒ Custom test suite
Troubleshoot and debug your device software using one or more prebuilt test cases.

Protocol [Info](#)

Choose the communication protocol used by your device. Tests for that particular protocol will be presented when you create your test suite.

☒ MQTT 3.1.1

☐ MQTT 5

☐ MQTT 3.1.1 over WebSocket

☐ MQTT 5 over WebSocket

Cancel **Next**

7. Drag all 7 TLS test cases into the “Test group 1” window. The completed group should look similar to the graphic below.

Test cases are the individual prebuilt test that are configured to test with things

Show all test cases ▼

▼ TLS (7)

|| TLS Connect

|| TLS Receive Maximum Size Fragments

|| TLS Support AWS IoT Cipher Suites

|| TLS Unsecure Server Cert

|| TLS Incorrect Subject Name Server Cert

|| TLS Large Size Server Cert

|| TLS Expired Server Cert

▼ MQTT (14)

|| MQTT Connect

|| MQTT Connect Jitter Retries

|| MQTT Connect Exponential Backoff Retries

|| MQTT Reconnect Backoff Retries On Server Disconnect

Drag the test cases over

Start

Starting point of this test suite.

All test groups and test cases will execute in descending order from this point. Even if an error occurs on a test case.

+ Add test group

Test group 1 Edit

|| TLS Connect Edit

|| TLS Receive Maximum Size Fragments Edit

|| TLS Support AWS IoT Cipher Suites Edit

|| TLS Unsecure Server Cert Edit

|| TLS Incorrect Subject Name Server Cert Edit

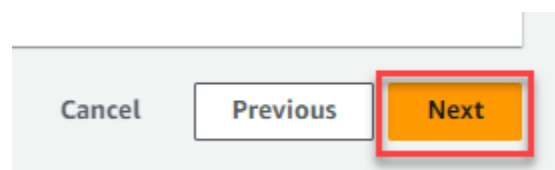
|| TLS Large Size Server Cert Edit

|| TLS Expired Server Cert Edit

When the tests in this group are completed, testing will continue with the next group.

+ Add test group

8. Scroll down and click the “Next” link



Next we need to create a Device Role for the device to use for the test connection

9. Fill out the form as shown below . . . If you used a different **ClientID** or **Topics**, use your values.
10. Select the “**Next**” link at the bottom of the page

Device role

Device role

Info

AWS IoT Core Device Advisor requires permission to perform AWS IoT MQTT actions on behalf of your test device.

☒ Create new role
Create and use a new device role

☐ Select an existing role
Use an existing device role

Role name

MyDeviceAdvisorRole

Permissions

Info

Choose which actions and the associated resources for AWS IoT Core Device Advisor to access using this role. You can enter a specific resource or resource prefix. To enter multiple values for a resource, use commas to separate the values. [Learn more](#)

Action	Resource type	Resource
☒ Connect	ClientId	BW-RASyn We support \$ and other special characters. * asterisk can only be added at the end
☒ Publish	Topic	MyTopic We support \$ and other special characters. * asterisk can only be added at the end
☒ Subscribe	TopicFilter	MyTopic We support \$ and other special characters. * asterisk can only be added at the end
☒ Receive	Topic	MyTopic We support \$ and other special characters. * asterisk can only be added at the end
☐ RetainPublish	Topic	<i>Specify topics to publish a retained message to, e.g. MyTopic, MyTopic</i> We support \$ and other special characters. * asterisk can only be added at the end

Cancel

Previous

Next

11. Your Test Suite Details are shown

12. Scroll down and click the “**Create test suite**” link at the bottom of the page

Review

Step 1: Test suite type

Edit

Test suite type

Custom test suite

Protocol

MQTT 3.1.1

Step 2: Test suite

Edit

Test suite details

Test suite name

Test suite June 26, 2024, 15:51:44 (UTC-0400)

Suite version

v1

Test type

Custom test suite

Start

Starting point of this test suite.

↓

Test group 1

TLS Connect

TLS Receive Maximum Size Fragments

TLS Support AWS IoT Cipher Suites

TLS Unsecure Server Cert

TLS Incorrect Subject Name Server Cert

TLS Large Size Server Cert

TLS Expired Server Cert

When the tests in this group are completed, testing will continue with the next group.

↓

End

End point of this test suite.

Step 3: Device role

Edit

Device role details

Device role type

Create new role

Device role name

MyDeviceAdvisorRole

Cancel

Previous

Create test suite

13. Your new Test Suite is listed in the Test Suite List

Test suites [Info](#)

Test suites (7) Find test suite Actions Create test suite

	Name	Test Type	Protocol	Date created
☐	Test suite June 26, 2024, 15:51:44 (UTC-0400)	Custom test suite	MQTT 3.1.1	June 26, 2024, 16:14:20 (UTC-0400)

Run the Test

1. From the “**Test suites**” list, select the Custom test suite we just created
2. From the “**Actions**” link at the top right, open the pull down menu and select “**Run test suite**”

[AWS IoT](#) > [Test](#) > [Device Advisor](#) > [Test suites](#) > Test suite June 26, 2024, 15:51:44 (UTC-0400)

Test suite June 26, 2024, 15:51:44 (UTC-0400)

Actions

- Run test suite
- Edit
- Delete

Test suite details

Created June 26, 2024, 16:14:20 (UTC-0400)	Test type Custom test suite	Protocol MQTT 3.1.1	Suite version v1
Suite definition ARN <code>arn:aws:iotdeviceadvisor:us-west-2:763534712565:suitedefinition/i4da89he03gj</code>		Device role ARN <code>arn:aws:iam::763534712565:role/service-role/MyDeviceAdvisorRole</code>	

Activity Log

3. The “Run configuration” page opens
 - a. Select “Things”
 - b. Select the device you created i.e., **BW-RASyn**
 - c. Select “Account-level endpoint”
 - d. Copy the test endpoint string. Use this string to update your config.ini:
[AWS]→Endpoint= configuration item.
 - e. **DON'T RUN THE TEST YET!!!!**

Run configuration

Select test devices

Select the IoT thing/certificate to test using the test suite. If not listed below, you must first create a thing/certificate registered with IoT Core before you can run the test suite.

☒ Things
Choose a thing for this test suite. To create a new thing, go to [IoT Things](#).

☐ Certificates
Choose a certificate for this test suite. To create a new certificate, go to [IoT Certificates](#).

Things (4)

☒ BW-RASyn

☐ testing

☐ AES-RASYNB-120-G

☐ RASynVideoSeries

Test endpoint

Choose the endpoint that best fits your situation. If you want to simultaneously run multiple test suites then use 'Device-level endpoint', if you want to run only one test suite at a time then choose the 'Account-level endpoint'.

☒ Account-level endpoint
Using this endpoint, you can only run one test suite at a time.

☐ Device-level endpoint
Using this endpoint, you can run multiple test suites simultaneously.

Copy and paste this endpoint to your test device.

t31p7o4mlgajx1.deviceadvisor.iot.us-west-2.amazonaws.com

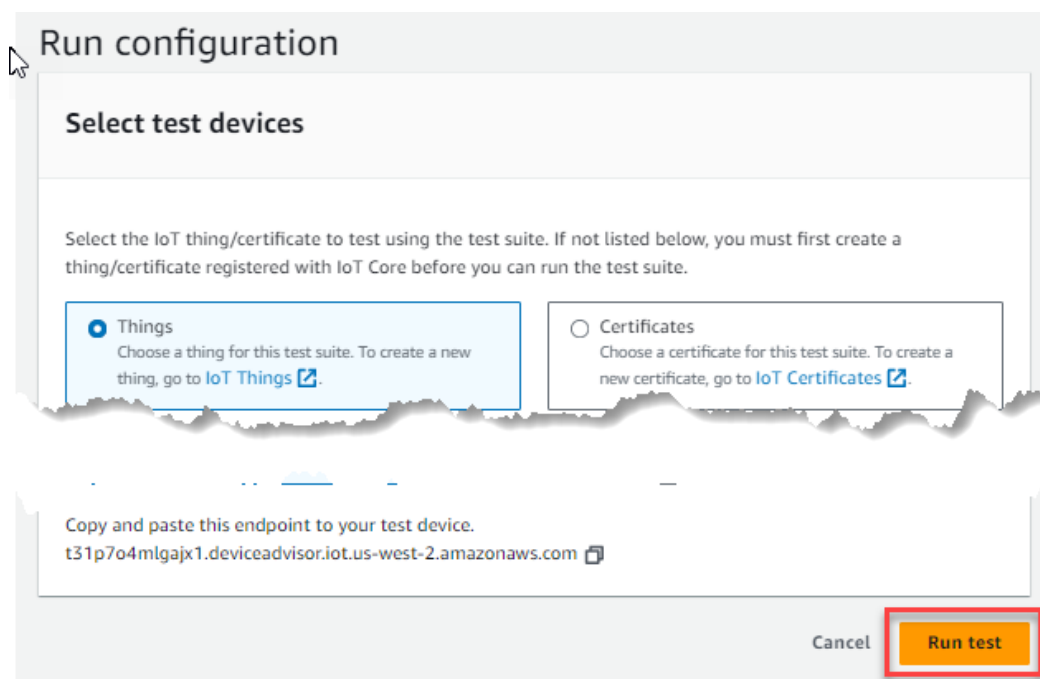
Cancel

Run test

4. Edit the config.ini file on your RASynBoard microSD card file to use the new endpoint string

```
[AWS] # http://avnet.me/RASynDocsAWSIoTCore
# Define the specific details to support connecting to AWS IoT Core.
# Instructions for setting up an AWS IoT Core connection: https://github.com/avnet/RASynDocsAWSIoTCore
Endpoint=t31p7o4mlgajx1.deviceadvisor.iot.us-west-2.amazonaws.com
Device_Unique_ID=BW-RASyn
MQTT_Pub_Topic=MyTopic
MQTT_Sub_Topic=MyTopic
```

5. Power cycle your RASynBoard to take the new configuration
6. Back in the AWS console web page, click the “Run test” button



Run configuration

Select test devices

Select the IoT thing/certificate to test using the test suite. If not listed below, you must first create a thing/certificate registered with IoT Core before you can run the test suite.

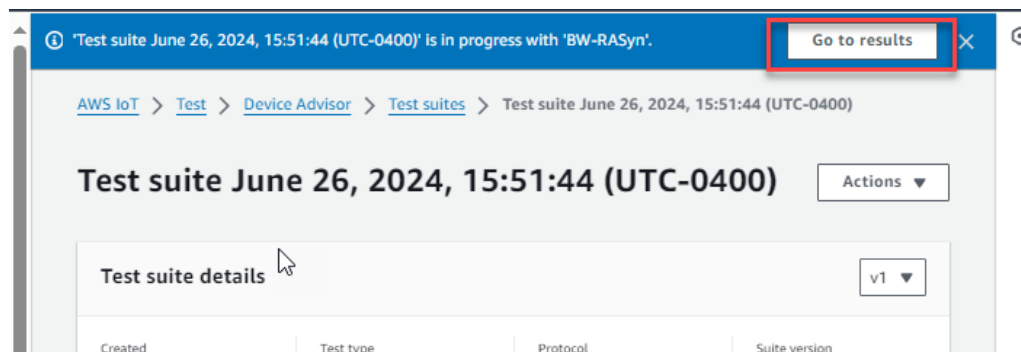
☒ Things
Choose a thing for this test suite. To create a new thing, go to [IoT Things](#).

☐ Certificates
Choose a certificate for this test suite. To create a new certificate, go to [IoT Certificates](#).

Copy and paste this endpoint to your test device.
t31p7o4mlgajx1.deviceadvisor.iot.us-west-2.amazonaws.com

Cancel Run test

7. The test starts and AWS Console shows a link at the top “Go to results”, click the link



Test suite June 26, 2024, 15:51:44 (UTC-0400) is in progress with 'BW-RASyn'.

Go to results

AWS IoT > Test > Device Advisor > Test suites > Test suite June 26, 2024, 15:51:44 (UTC-0400)

Test suite June 26, 2024, 15:51:44 (UTC-0400) Actions

Test suite details v1

Created	Test type	Protocol	Suite version
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8. The “Test runs and results” page opens

Test runs and results

Summary

Number of IoT things available	Number of IoT certificates available	Number of test suites running
4	9	1

Results of test runs (in progress and completed)

< 1 2 3 4 > ⚙

Name	Timestamp	Suite version	Status	Passed	Failed	Duration
Test suite June 26, 2024, 15:51:44 (UTC-0400)	June 27, 2024, 14:18:40 (UTC-0400)	v1	🔄 In Progress	-	-	-
Failing TLS test suite	June 04, 2024, 16:02:01 (UTC-0400)	v1	❌ Failed	-	3	6 minutes
Failing TLS test suite	June 04, 2024, 14:40:50 (UTC-0400)	v1	❌ Failed	1	2	5 minutes
Failing TLS test suite	June 04, 2024, 14:28:53 (UTC-0400)	v1	❌ Failed	1	2	4 minutes

9. Click on the test run link that’s in progress

10. This page shows the tests and the results as they complete

June 27, 2024, 14:18:40 (UTC-0400) [Test suite log](#) Actions ▼

Summary

Device BW-RASyn	Protocol MQTT 3.1.1	Suite version v1	Created June 27, 2024, 14:18:40 (UTC-0400)	Status 🔄 In Progress
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Test group 1 (7) 🔄 In Progress

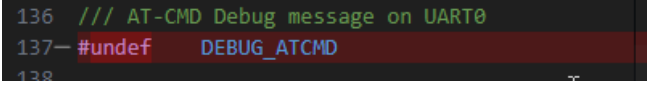
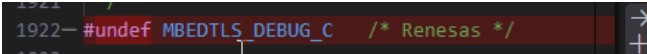
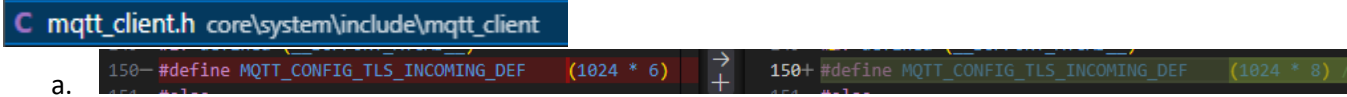
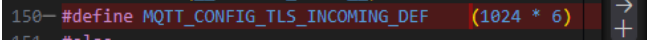
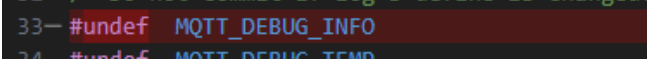
Test	Result	System message	Logs
TLS Connect	✅ Passed	No issues found	Test case log
TLS Receive Maximum Size Fragments	✅ Passed	No issues found	Test case log
TLS Support AWS IoT Cipher Suites	⚠️ Passed	FOUND_UNSUPPORTED_CIPHER_SUITE	Test case log
TLS Unsecure Server Cert	🔄 In Progress		Test case log
TLS Incorrect Subject Name Server Cert	⌚ Pending		
TLS Large Size Server Cert	⌚ Pending		
TLS Expired Server Cert	⌚ Pending		

NOTE: Depending on what the test is doing, you may need to power cycle the RASynBoard to kick off a new TLS connection. You can also force the RASynBoard to try and send telemetry by triggering an inference event, speak “up”. This will cause the application to check the MQTT connection status and if not connected, it will roll the state machine back and try to reestablish the MQTT over TLS connection.

11. Eventually all tests will complete showing the results

DA16600 Code Changes

This section identifies changes I've already made in the DA16600 AT-CMD code base in this effort.

1. 
a. 
b. For ATCMD debug
2. 
a. 
b. For TLS debug
3. 
a. 
b. Required for the large certificate test to pass
4. 
a. 
b. For MQTT debug

Revision History

Revision Number	Date	Note
V1	6/27/24	Initial document